

DETAILED DESCRIPTION OF THE INVENTION — SENSOR PLACEMENT TOPOLOGIES AND SPATIAL MEASUREMENT ARCHITECTURES

The intraluminal physiologic monitoring device may employ a wide variety of sensor placement configurations designed to capture spatially distributed physiologic information. Sensor placement topology may significantly influence measurement fidelity, anatomical coverage, calibration behavior, and derived physiologic interpretation.

In various embodiments, sensing elements may be arranged circumferentially around the device body forming one or more sensing rings configured to measure radial tissue interaction. Multiple rings may be positioned along the longitudinal axis to enable spatial mapping of physiologic conditions across different regions of the monitored lumen.

Longitudinal sensing arrangements may include linear sensor strips extending along the device length. Such configurations may allow detection of gradients, directional changes, insertion depth estimation, or localized physiologic variation across anatomical zones.

Segmented sensing architectures may divide the device into discrete sensing regions, each independently measuring pressure, impedance, dielectric response, temperature, or motion characteristics. Segmentation may improve resolution of localized physiologic responses and enable adaptive regional analysis.

Matrix-based sensor arrays may be implemented using flexible circuit substrates or distributed electrode networks forming two-dimensional sensing grids. Matrix sensing may allow reconstruction of spatial interaction patterns between tissue and device surfaces.

Sensors may be embedded beneath outer materials, integrated within compliant layers, exposed through sensing windows, or positioned within deformable structures that mechanically couple tissue interaction to sensing elements. Protective coatings, isolation layers, or compliant interfaces may be used to optimize durability and signal quality.

Hybrid sensor placement strategies may combine circumferential, longitudinal, and distributed sensing approaches within a single device architecture. Sensor activation may occur simultaneously, sequentially, adaptively, or in response to detected physiologic events.

Certain embodiments may incorporate movable, reconfigurable, or dynamically addressable sensing elements allowing adaptive sensing regions or variable spatial resolution during operation. Future implementations may include soft robotic positioning systems or electronically selectable sensing zones.

The disclosure encompasses any spatial arrangement of sensors capable of measuring physiologic interaction within an intraluminal environment. Variations in sensor number, placement density, orientation, or activation strategy are considered within the scope of the invention, including future sensing architectures not yet commercially realized.